

Solving linear Homogeneous Recurrences.

$$\begin{aligned} a_n &= a_{n-1} + 6a_{n-2} \\ a_0 &= 1, a_1 = 8 \end{aligned}$$

- ① characteristic equation of recurrence
- ② Find roots of charac. equ. (factoring)
- ③ Determine general form of non-recursive solution
- ④ Find coefficients of general form (system of equations)

$$\textcircled{1} \quad a_n - a_{n-1} - 6a_{n-2} = 0$$

$$\underbrace{x^n - x^{n-1} - 6x^{n-2}}_{\downarrow \div x^{n-2}} = 0$$

$$x^2 - x - 6 = 0 \quad \text{characteristic equation}$$

$$4a_n - 6a_{n-1} + 8a_{n-3} = 0$$

$$4x^n - 6x^{n-1} + 8x^{n-3} = 0$$

$$\downarrow \div x^{n-3}$$

$$4x^3 - 6x^2 + 8 = 0$$

Charact. equ.

$$\textcircled{2} \quad x^2 - x - 6 = 0$$

factoring: $(x-3)(x+2) = 0$ $\begin{cases} x_1 = 3 \\ x_2 = -2 \end{cases}$ roots

find roots of degree-2 equ.

$$ax^2 + bx + c = 0$$

$$x_1, x_2 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\rightarrow x_1, x_2 = \frac{1 \pm \sqrt{1 - (-24)}}{2} \quad \begin{cases} x_1 = \frac{1 + \sqrt{25}}{2} = 3 \\ x_2 = \frac{1 - \sqrt{25}}{2} = -2 \end{cases}$$

$$\textcircled{3} \quad a_n = \alpha (x_1)^n + \beta (x_2)^n \quad \alpha, \beta: \text{numbers}$$

$$a_n = \alpha (3)^n + \beta (-2)^n$$

$$\textcircled{4} \quad \begin{aligned} a_0 = 1 &\Rightarrow a_0 = \alpha (3)^0 + \beta (-2)^0 = 1 \Rightarrow \alpha + \beta = 1 \\ a_1 = 8 &\Rightarrow a_1 = \alpha (3)^1 + \beta (-2)^1 = 8 \Rightarrow 3\alpha - 2\beta = 8 \end{aligned}$$

$$\begin{aligned} \alpha + \beta &= 1 \\ 3\alpha - 2\beta &= 8 \end{aligned} \quad \left\{ \begin{array}{l} \times 2 \\ \end{array} \right. \quad \begin{aligned} 2\alpha + 2\beta &= 2 \\ 3\alpha - 2\beta &= 8 \end{aligned} \quad +$$

$$5\alpha = 10$$

$$\downarrow$$

$$\boxed{\alpha = 2}$$

$$\boxed{\beta = -1}$$

$$a_n = 2(3)^n + (-1)(-2)^n \quad \text{final answer}$$

$$a_2 = a_1 + 6a_0 = 8 + 6 \times 1 = 14 \quad (\text{recursive formula})$$

$$a_2 = 2(3)^2 + (-1)(-2)^2 = 18 - 4 = 14 \quad (\text{non-rec. formula})$$

$$a_n = 4a_{n-1} - 4a_{n-2}, \quad n \geq 2$$

$$a_0 = 1, \quad a_1 = 3$$

$$\textcircled{1} \quad a_n - 4a_{n-1} + 4a_{n-2} = 0$$

$$x^n - 4x^{n-1} + 4x^{n-2} = 0$$

$$\downarrow \div x^{n-2}$$

$$\bullet x^2 - 4x + 4 = 0 \quad \text{charac. equ.}$$

$$\downarrow$$

$$\textcircled{2} \quad (x-2)(x-2) = 0 \quad \begin{cases} x_1 = 2 \\ x_2 = 2 \end{cases}$$

$$x_1, x_2 = \frac{4 \pm \sqrt{16-16}}{2} \quad \begin{cases} x_1 = \frac{4 + \sqrt{0}}{2} = 2 \\ x_2 = \frac{4 - \sqrt{0}}{2} = 2 \end{cases}$$

$$\textcircled{3} \quad a_n = \alpha (2)^n + \beta n (2)^n \quad \text{because the roots are equal}$$

$$\uparrow n!$$

$$\text{assume: } \underline{3}, \underline{2}, \underline{2}, \underline{2}, 8, 8, 8, 8$$

$$\Rightarrow a_n = \alpha \cdot n! (3^n) + \beta \cdot n! (2)^n + \beta \cdot n! (2)^n + \beta \cdot n! (2)^n + \gamma \cdot n! (8)^n + \gamma \cdot n! (8)^n + \gamma \cdot n! (8)^n + \gamma \cdot n! (8)^n$$

$$\textcircled{4} \quad \text{find } \alpha, \beta$$

$$a_0 = 1 \Rightarrow a_0 = \alpha (2)^0 + \beta \times 0 \times (2)^0 = 1 \Rightarrow \boxed{\alpha = 1}$$

$$a_1 = 3 \Rightarrow a_1 = \alpha (2)^1 + \beta \times 1 \times (2)^1 = 3 \Rightarrow 2\alpha + 2\beta = 3$$

$$\downarrow$$

$$2 + 2\beta = 3$$

$$2\beta = 1$$

$$\boxed{\beta = \frac{1}{2}}$$

$$a_n = 2^n + \frac{1}{2} n 2^n$$

$$= 2^n + 2^{n-1} n$$

$$= 2^{n-1} (2+n)$$

$$\begin{aligned} &\rightarrow a_2 = 4a_1 - 4a_0 = 4 \times 3 - 4 \times 1 = 8 \quad \text{rec.} \\ &\rightarrow a_2 = 2^{2-1} (2+2) = 2(4) = 8 \quad \text{non-rec.} \end{aligned}$$

Fibonacci

$$a_0 = 0, a_1 = 1, a_n = a_{n-1} + a_{n-2}$$

① $a_n - a_{n-1} - a_{n-2} = 0$ optional

$\rightarrow x^n - x^{n-1} - x^{n-2} = 0 \quad \div x^{n-2}$

$$\overset{a}{x^2} - \overset{b}{x} - \overset{c}{1} = 0$$

② $x_1, x_2 = \frac{1 \pm \sqrt{1 - (-4)}}{2} \quad \left\{ \begin{array}{l} x_1 = \frac{1 + \sqrt{5}}{2} \\ x_2 = \frac{1 - \sqrt{5}}{2} \end{array} \right.$

③ $a_n = \alpha \left(\frac{1 + \sqrt{5}}{2} \right)^n + \beta \left(\frac{1 - \sqrt{5}}{2} \right)^n$

④ Find α, β

$$a_0 = 0 \Rightarrow a_0 = \alpha \left(\frac{1 + \sqrt{5}}{2} \right)^0 + \beta \left(\frac{1 - \sqrt{5}}{2} \right)^0 = 0 \Rightarrow \alpha + \beta = 0 \Rightarrow \beta = -\alpha$$

$$a_1 = 1 \Rightarrow a_1 = \alpha \left(\frac{1 + \sqrt{5}}{2} \right)^1 + \beta \left(\frac{1 - \sqrt{5}}{2} \right)^1 = 1$$

$$\alpha \left(\frac{1 + \sqrt{5}}{2} \right) - \alpha \left(\frac{1 - \sqrt{5}}{2} \right) = 1$$

$$\frac{\alpha}{2} [(1 + \sqrt{5}) - (1 - \sqrt{5})] = 1$$

$$\frac{\alpha}{2} [2\sqrt{5}] = 1 \rightarrow \boxed{\alpha = \frac{1}{\sqrt{5}}} \quad , \quad \boxed{\beta = \frac{-1}{\sqrt{5}}}$$

$$a_n = \left(\frac{1}{\sqrt{5}} \right) \left(\frac{1 + \sqrt{5}}{2} \right)^n + \left(\frac{-1}{\sqrt{5}} \right) \left(\frac{1 - \sqrt{5}}{2} \right)^n$$

Find a non-recursive formula for the number of binary strings of size n with no consecutive 0. (a_n)

$n=1$ 1, 0 $a_1 = 2$

$n=2$ 11, 10, 01, ~~00~~ $a_2 = 3$

$n=3$ 111, 110, 101, 011, ~~100~~, ~~010~~, ~~001~~, ~~000~~ $a_3 = 5$

⋮

$a_n = ?$

We find a recursive formula and

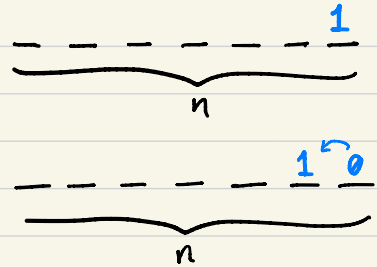
then solve it

$$a_n = a_n^{(1)} + a_n^{(0)}$$

↙ end with 1
↙ end with 0

$\downarrow$
 a_{n-1}

$\downarrow$
 a_{n-2}



$$a_n = a_{n-1} + a_{n-2}$$

$$a_1 = 2$$

$$a_2 = 3$$

→ solve it

Similar to Fibonacci